

Arithmetic Power Geometry Descriptors for Interpretable Structural Analysis of Single-Cell PBMC Transcriptomes

Mohammad Amir Khusru Akhtar

Usha Martin University, Ranchi-834001, Jharkhand, India

akakhtar.2024@gmail.com

Abstract

Single-cell RNA sequencing produces high-dimensional molecular profiles, whereas compact cell-level summaries often reduce each transcriptome to one diversity or concentration statistic. We developed an Arithmetic Power Geometry (APG) representation that maps each nonnegative expression vector to normalized squared weights and then measures entropy, concentration, effective dimension, a bounded exponent-deformation curve, and integrated deformation. The method was evaluated on the official 10x Genomics Fresh 68k peripheral blood mononuclear cell dataset from Donor A, comprising 68,579 cells. APG descriptors were compared directly with Shannon entropy, Rényi-2 entropy, Simpson diversity, Gini inequality, Herfindahl concentration, participation ratio, and detected-gene count. Marker-supported scoring yielded confident provisional annotations for 56,903 cells across eight immune or blood-associated classes, while 11,676 cells were retained as uncertain. Programme-level APG was computed across nine curated immune and cellular programmes. In a balanced 6,400-cell internal benchmark with identical three-fold stratified splits, classical diversity descriptors achieved macro-F1 0.3027 ± 0.0058 ; adding gene-level APG increased macro-F1 to 0.3792 ± 0.0104 . The combined classical, gene-level APG, and programme-level APG representation achieved 0.5299 ± 0.0037 macro-F1 and 0.5400 ± 0.0020 balanced accuracy. Programme APG maximum concentration, integrated deformation, and $D(3)$ showed the largest between-class effect sizes ($\epsilon^2 = 0.563, 0.541, \text{ and } 0.526$, respectively). Under binomial count thinning to 20% of the original depth, normalized APG entropy, $D(3)$, and integrated deformation retained Spearman correlations of 0.917, 0.878, and 0.893 with their full-depth values. All tested compact spaces nevertheless had negative silhouette coefficients, and the classification labels were derived computationally from marker modules. The results therefore establish a reproducible one-donor methodological benchmark rather than external biological or clinical validation. APG contributes a multi-exponent structural profile that complements, but does not replace, established single-cell representations and diversity measures.

Keywords: Arithmetic Power Geometry; single-cell RNA sequencing; PBMC; entropy; exponent deformation; gene programmes; interpretable representation; transcriptomic diversity

1 Introduction

Single-cell RNA sequencing (scRNA-seq) enables transcriptomic measurement at the resolution of individual cells and has become central to the study of cellular heterogeneity. Droplet-based single-cell sequencing technologies have made it practical to profile tens of thousands of cells in a single experiment, including the widely used 68k peripheral blood mononuclear cell (PBMC) benchmark dataset originally described by [Zheng et al. \(2017\)](#). Contemporary workflows integrate quality control, normalization, highly variable gene selection, dimensionality

reduction, clustering, cell-type annotation, differential expression analysis, and increasingly multimodal data integration (Luecken and Theis, 2019; Heumos et al., 2023). Scalable software such as Scanpy and Seurat supports these workflows (Wolf et al., 2018; Stuart et al., 2019; Hao et al., 2021), while probabilistic latent-variable models such as scVI explicitly model technical variation and biological uncertainty (Lopez et al., 2018).

These methods preserve substantial transcriptomic information, but their latent coordinates are not generally reducible to a small set of biologically transparent structural quantities. Conversely, compact statistics such as Shannon entropy, Rényi entropy, Simpson diversity, concentration, inequality, and effective dimension are interpretable but describe only particular properties of a distribution (Shannon, 1948; Rényi, 1961; Simpson, 1949; Gini, 1912; Lloyd and Ghelardi, 1964). A single entropy value, for example, does not explicitly describe how the distribution responds when dominant coordinates are progressively amplified.

Arithmetic Power Geometry (APG) treats an exponent as a deformation parameter and associates the local response near the Euclidean exponent with the entropy of normalized squared weights (Akhtar, 2026a,b). This suggests a cell-level representation in which a transcriptome is not described only by a static diversity value, but by a bounded curve showing how its distributed expression mass changes under exponent deformation. The intended contribution is not to rename Shannon entropy or established concentration indices. It is to test whether a multi-exponent APG profile and its integrated deformation contribute information beyond those established summaries.

Single-cell gene-set and regulatory-programme scoring already provides a biologically interpretable intermediate representation. SCENIC, UCell, and related approaches quantify regulatory or functional signatures at single-cell resolution (Aibar et al., 2017; Andreatta and Carmona, 2021). Recent benchmarks show that pathway scores are affected by gene-set size, sparsity, noise, and scoring design (Wang et al., 2024); marker-gene selection and cell annotation likewise remain method-dependent (Abdelaal et al., 2019; Pullin et al., 2024). For that reason, the present study evaluates APG both at the gene level and over a deliberately small set of curated immune and cellular programmes, while treating the resulting labels as provisional computational annotations.

The study addresses four questions. First, can APG descriptors be calculated reproducibly for a large sparse scRNA-seq matrix? Second, do gene-level APG quantities show between-class structure beyond classical diversity statistics? Third, does programme-level APG provide additional compact discrimination? Fourth, how stable are APG descriptors when UMI counts are reduced by controlled thinning? The answers are evaluated on one canonical donor dataset, with explicit limits on generalization.

2 Mathematical framework

2.1 Classical cell-level distributions

For cell c , let $z_{ci} \geq 0$ denote the library-normalized and log-transformed expression of gene i , $i = 1, \dots, m$. The ordinary expression weights are

$$q_{ci} = \frac{z_{ci}}{\sum_{j=1}^m z_{cj}}, \quad \sum_i q_{ci} = 1. \quad (1)$$

Classical descriptors were computed from $\mathbf{q}_c$, including Shannon entropy $H_q = -\sum_i q_{ci} \log q_{ci}$, Rényi-2 entropy $H_2 = -\log \sum_i q_{ci}^2$, Simpson diversity $1 - \sum_i q_{ci}^2$, Herfindahl concentration $\sum_i q_{ci}^2$, participation ratio $(\sum_i q_{ci}^2)^{-1}$, and Gini inequality.

2.2 APG weights and descriptors

The APG Euclidean weights are

$$w_{ci} = \frac{z_{ci}^2}{\sum_{j=1}^m z_{cj}^2}, \quad \sum_i w_{ci} = 1. \quad (2)$$

The associated APG entropy and normalized entropy are

$$H_c^{\text{APG}} = -\sum_i w_{ci} \log w_{ci}, \quad \tilde{H}_c^{\text{APG}} = \frac{H_c^{\text{APG}}}{\log m}. \quad (3)$$

Maximum and quadratic concentration are $C_{\max,c} = \max_i w_{ci}$ and $C_{2,c} = \sum_i w_{ci}^2$. Effective dimension and participation ratio are $\exp(H_c^{\text{APG}})$ and $C_{2,c}^{-1}$, respectively.

For exponent $p \geq 2$, the APG defect is

$$D_c(p) = 1 - \sum_i w_{ci}^{p/2}. \quad (4)$$

At $p = 2$, $D_c(2) = 0$. For $p > 2$, $D_c(p)$ is bounded in $[0, 1)$ and is nondecreasing in p . Near $p = 2$,

$$D_c(2 + \varepsilon) = \frac{\varepsilon}{2} H_c^{\text{APG}} + O(\varepsilon^2), \quad (5)$$

so APG entropy controls the initial deformation rate. The local expansion in [equation \(5\)](#) motivates the deformation summary used here. The integrated deformation used here is

$$A_c(2, 6) = \int_2^6 D_c(p) dp, \quad (6)$$

approximated by trapezoidal quadrature over $p \in \{2, 2.25, 2.5, 3, 4, 5, 6\}$.

2.3 Programme-level APG

Let $s_{ck} \geq 0$ be the mean normalized expression score for programme k , $k = 1, \dots, K$. Programme APG uses

$$u_{ck} = \frac{s_{ck}^2}{\sum_{\ell=1}^K s_{c\ell}^2} \quad (7)$$

and applies [equations \(3\)](#), [\(4\)](#) and [\(6\)](#) to $\mathbf{u}_c$. This layer quantifies whether the programme mass is balanced across multiple functions or concentrated in a small number of programmes.

3 Materials and methods

3.1 Dataset and preprocessing

The analysis used the official filtered gene–cell matrix for Fresh 68k PBMCs (Donor A). The Cell Ranger report recorded 68,579 cells, 20,491 mean reads per cell, 525 median detected genes per cell, 1,292 median UMI counts per cell, 96.6% reads in cells, and 74.2% reads mapped confidently to the transcriptome. The reference genome was hg19 and Cell Ranger version 1.1.0. The matrix contained 32,738 genes. The complete dataset summary is provided in [table 2](#).

For each cell, UMI counts were divided by the library total, multiplied by 10,000, and transformed by $\log(1 + x)$. Classical weights used [equation \(1\)](#); APG weights used [equation \(2\)](#). Calculations were performed directly on sparse matrices. The evaluated descriptors are summarized in [table 1](#).

Table 1: Cell-level descriptors evaluated in the benchmark.

Descriptor	Definition	Interpretation
Shannon entropy	$-\sum_i q_i \log q_i$	Breadth of ordinary normalized expression
Rényi-2 entropy	$-\log \sum_i q_i^2$	Diversity weighted toward dominant genes
Simpson diversity	$1 - \sum_i q_i^2$	Probability-style diversity measure
Herfindahl concentration	$\sum_i q_i^2$	Ordinary expression concentration
Participation ratio	$(\sum_i q_i^2)^{-1}$	Effective number under quadratic concentration
Gini inequality	Rank-weighted expression inequality	Unevenness across the complete gene set
APG normalized entropy	$H^{\text{APG}} / \log m$	Entropy of squared expression weights
APG $D(p)$	$1 - \sum_i w_i^{p/2}$	Bounded exponent-deformation profile
APG integrated deformation	$\int_2^6 D(p) dp$	Accumulated deformation over the tested range
Programme APG	APG applied to programme scores	Balance and deformation across biological programmes

Table 2: Dataset and sequencing summary from the official Cell Ranger report.

Measure	Value
Cells	68,579
Genes in filtered matrix	32,738
Mean reads per cell	20,491
Median genes per cell	525
Median UMI counts per cell	1,292
Fraction of reads in cells	96.6%
Valid barcodes	95.2%
Reads mapped confidently to transcriptome	74.2%
Reference genome	hg19
Cell Ranger version	1.1.0

3.2 Marker-supported provisional annotations

Eight marker modules represented CD4 T cells, CD8 T cells, natural killer (NK) cells, B cells, CD14 monocytes, FCGR3A monocytes, dendritic cells, and platelets. Each module score was the mean normalized expression of its available markers. Scores were standardized within modules across all cells. A cell was assigned to the module with the highest standardized score only when the top score was at least 0.25 and its margin over the second-highest score was at least 0.10; otherwise it was labelled uncertain. This deterministic procedure retained uncertainty rather than forcing every cell into a class. Because marker selection and thresholding can affect annotation (Aran et al., 2019; Abdelaal et al., 2019; Pullin et al., 2024), the labels were used only for internal methodological evaluation.

3.3 Curated biological programmes

Nine programmes were scored: T-cell receptor signalling, cytotoxicity, B-cell receptor activity, interferon response, inflammatory myeloid activity, oxidative phosphorylation, ribosomal activity, cell cycle, and apoptosis/stress. Each score was the mean normalized expression of a short curated marker set, followed by programme APG construction using [equation \(7\)](#). The complete programme and annotation marker definitions are reported in [tables 9](#) and [10](#). These scores are transparent compact modules, not a claim of pathway-enrichment equivalence to AUCell, UCell, or other established gene-set methods ([Aibar et al., 2017](#); [Andreatta and Carmona, 2021](#); [Wang et al., 2024](#)).

3.4 Statistical comparisons

For each descriptor, global differences across the eight confident provisional classes were evaluated with the Kruskal–Wallis test ([Kruskal and Wallis, 1952](#)). Effect size was estimated as

$$\epsilon^2 = \max \left\{ 0, \frac{H - k + 1}{N - k} \right\}, \quad (8)$$

where H is the Kruskal–Wallis statistic, k the number of groups, and N the total number of cells.

3.5 Internal classification benchmark

The classification analysis used a balanced subset of 6,400 confidently annotated cells, corresponding to 800 cells per provisional class. Five representations were compared: Shannon alone (M1); seven classical descriptors (M2); classical plus gene-level APG (M3); programme APG alone (M4); and classical plus gene- and programme-level APG (M5). Features were standardized within each training fold. Class-weighted multinomial logistic regression was evaluated with identical three-fold stratified splits. Macro-F1 and balanced accuracy were averaged over folds. The purpose was to measure incremental internal discrimination, not to establish external cell-type annotation performance. Scikit-learn was used for modeling ([Pedregosa et al., 2011](#)).

3.6 Silhouette analysis

Euclidean silhouette coefficients were computed on standardized descriptor spaces using a common sample of 1,500 cells ([Rousseeuw, 1987](#)). Positive values indicate that cells are, on average, closer to their own class than to neighboring classes; negative values indicate overlap or local assignment inconsistency.

3.7 Count-depth robustness

A fixed sample of 3,000 cells was subjected to binomial thinning of raw UMI counts, retaining 80%, 60%, 40%, or 20% of counts. The complete normalization and descriptor pipeline was repeated at each depth. Spearman correlation with each cell’s full-depth descriptor was used to measure rank stability.

Table 3: Provisional marker-supported annotation counts. Percentages use all 68,579 cells as the denominator.

Provisional class	Cells	Percentage
CD4 T	22,764	33.2%
CD8 T	10,482	15.3%
NK	9,325	13.6%
B cell	4,900	7.1%
FCGR3A monocyte	3,311	4.8%
CD14 monocyte	2,280	3.3%
Dendritic	2,148	3.1%
Platelet	1,693	2.5%
Uncertain	11,676	17.0%

4 Results

4.1 Annotation coverage and class composition

The confidence rule assigned 56,903 of 68,579 cells (83.0%) to one of eight provisional classes and retained 11,676 cells (17.0%) as uncertain. CD4 T cells constituted the largest confident group (22,764 cells), followed by CD8 T cells (10,482) and NK cells (9,325). The complete class distribution is reported in [table 3](#). Median marker-module patterns were consistent with the computational labels: the B-cell module was highest in B cells, the cytotoxic/NK module in NK cells, the CD14-myeloid module in CD14 monocytes, and the FCGR3A module in FCGR3A monocytes. These patterns support internal coherence, but the labels were not independently expert-reviewed.

4.2 Programme-level APG produced the largest between-class effects

All tested descriptors differed globally across the eight confident classes at machine-level underflow values of p (reported conservatively as $p < 10^{-300}$). Because the cell count is large, effect sizes are more informative than significance alone. The four largest effects were programme APG maximum concentration ($\epsilon^2 = 0.563$), programme integrated deformation ($\epsilon^2 = 0.541$), programme $D(3)$ ($\epsilon^2 = 0.526$), and programme normalized entropy ($\epsilon^2 = 0.493$). The largest gene-level APG effect was $D(6)$ ($\epsilon^2 = 0.169$). By comparison, Shannon entropy, Rényi-2 entropy, Simpson diversity, Herfindahl concentration, and participation ratio each had $\epsilon^2 \approx 0.081$. The ten largest effects are shown in [figure 1](#) and [table 4](#).

Gene-level $D(6)$ values were numerically close to one because the high exponent strongly suppresses all but the largest squared weights. Even within this compressed range, the class medians and distributions differed ([figure 2](#)). CD14 monocytes had the lowest median gene-level $D(6)$ among the provisional classes, consistent with comparatively greater concentration in dominant expression coordinates. The deformation curves in [figure 3](#) show rapid saturation after $p \approx 3$, indicating that lower exponents retain more dynamic range for high-dimensional gene-level profiles.

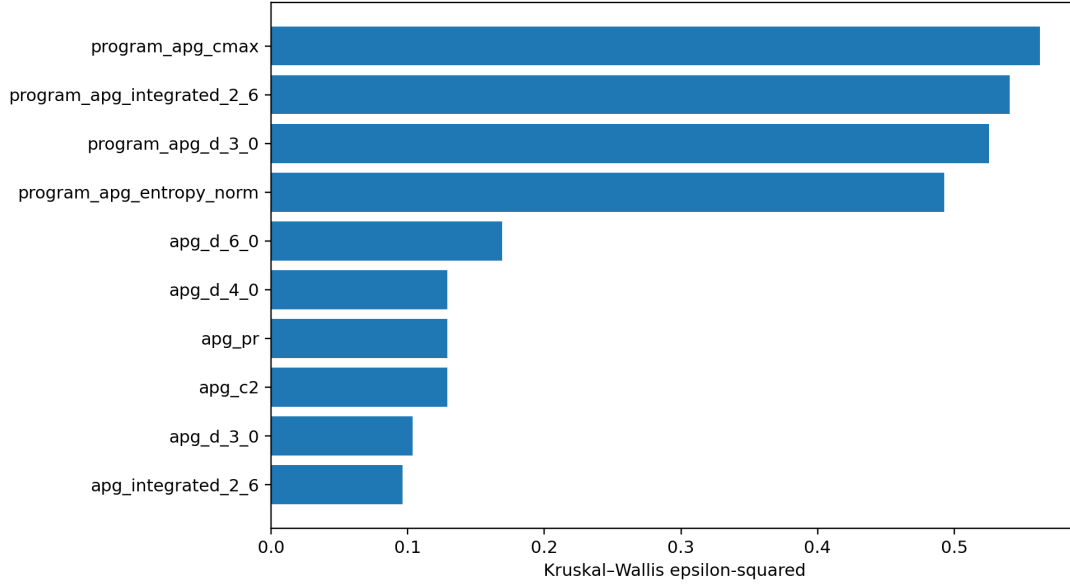


Figure 1: Largest Kruskal–Wallis epsilon-squared effect sizes across eight provisional PBMC classes. Programme-level APG descriptors produced the four largest effects.

Table 4: Ten largest global descriptor effects across 56,903 confidently annotated cells and eight groups.

Descriptor	Kruskal–Wallis H	p	ϵ^2
program apg cmax	32,030.3	$< 10^{-300}$	0.563
program apg integrated 2 6	30,770.5	$< 10^{-300}$	0.541
program apg d 3 0	29,908.2	$< 10^{-300}$	0.526
program apg entropy norm	28,042.3	$< 10^{-300}$	0.493
apg d 6 0	9,635.1	$< 10^{-300}$	0.169
apg pr	7,345.0	$< 10^{-300}$	0.129
apg d 4 0	7,345.0	$< 10^{-300}$	0.129
apg c2	7,345.0	$< 10^{-300}$	0.129
apg d 3 0	5,891.1	$< 10^{-300}$	0.103
apg integrated 2 6	5,488.0	$< 10^{-300}$	0.096

4.3 Programme deformation retained interpretable class structure

Programme integrated deformation showed visibly different distributions across the provisional classes (figure 4). NK cells had the highest median programme integrated deformation (2.010), followed by CD14 monocytes (1.866) and B cells (1.800). CD4 T cells and platelets had lower medians (0.789 and 0.903). Because integrated deformation increases when programme weights are less concentrated at the tested exponents, these results describe different organizations of the nine selected programmes rather than a universal ordering of biological complexity. Selected cell-type medians are reported in table 5.

4.4 APG features added internal discrimination beyond classical descriptors

The Shannon-only model obtained macro-F1 0.1710 ± 0.0057 and balanced accuracy 0.2189 ± 0.0040 . The seven-descriptor classical model increased macro-F1 to 0.3027 ± 0.0058 . Adding gene-level APG descriptors produced 0.3792 ± 0.0104 , an absolute increase of 0.0765 over the classical model. Programme APG alone achieved 0.3887 ± 0.0034 . The combined M5 representation

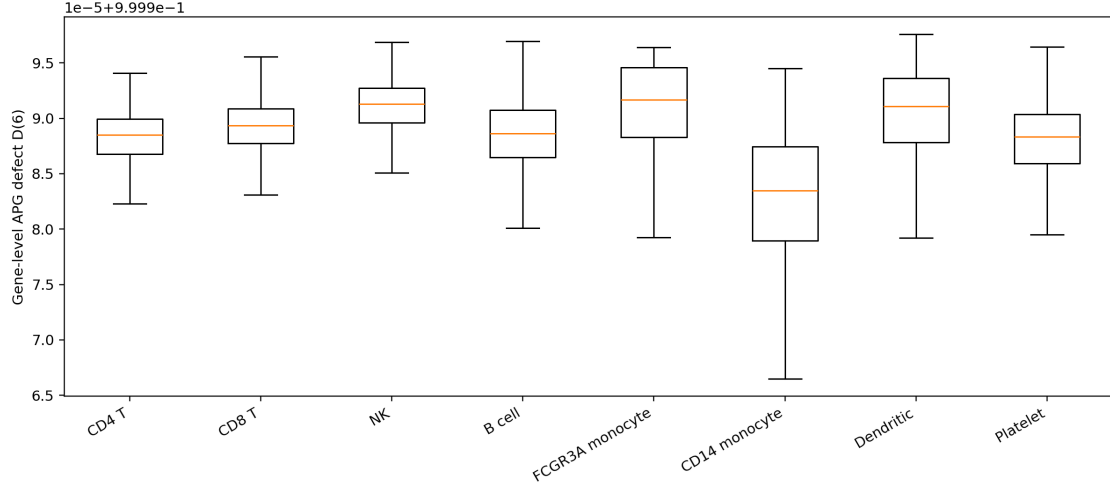


Figure 2: Distribution of gene-level APG defect $D(6)$ by provisional cell class. The offset notation on the axis reflects the narrow range close to one.

Table 5: Selected median descriptors by provisional class.

Class	n	Shannon	APG entropy norm.	APG $D(6)$	Programme $A(2,6)$
CD4 T	22,764	6.253	0.587	0.999988	0.789
CD8 T	10,482	6.225	0.587	0.999989	1.351
NK	9,325	6.243	0.591	0.999991	2.010
B cell	4,900	6.181	0.582	0.999988	1.800
FCGR3A monocyte	3,311	6.401	0.601	0.999992	1.730
CD14 monocyte	2,280	5.757	0.548	0.999983	1.866
Dendritic	2,148	6.323	0.596	0.999991	1.383
Platelet	1,693	6.238	0.587	0.999989	0.903

reached 0.5299 ± 0.0037 macro-F1 and 0.5400 ± 0.0020 balanced accuracy (table 6 and figure 5). Relative to M2, the M5 macro-F1 increase was 0.2272. These values show incremental consistency with the marker-derived classes within the same donor and analysis pipeline. They do not constitute independent annotation accuracy because the programme features and provisional labels draw on partially overlapping immune-marker information.

4.5 Compact descriptor spaces did not form well-separated Euclidean clusters

All silhouette coefficients were negative (table 7). M5 had the least negative value (-0.096), followed by M3 (-0.143). The programme-only representation had the lowest score (-0.214). This failure analysis is important: the descriptors improved supervised internal discrimination but did not yield compact, naturally separated Euclidean clusters. The result is compatible with overlapping immune states, provisional marker-derived labels, and the severe compression imposed by a small descriptor set.

4.6 APG descriptors remained rank-stable under count thinning

At 80% retained counts, all reported correlations exceeded 0.978. At 40% retained counts, normalized APG entropy, $D(3)$, and integrated deformation retained correlations of 0.966, 0.946, and 0.954, respectively. At 20%, these values were 0.917, 0.878, and 0.893 (table 8 and figure 6).

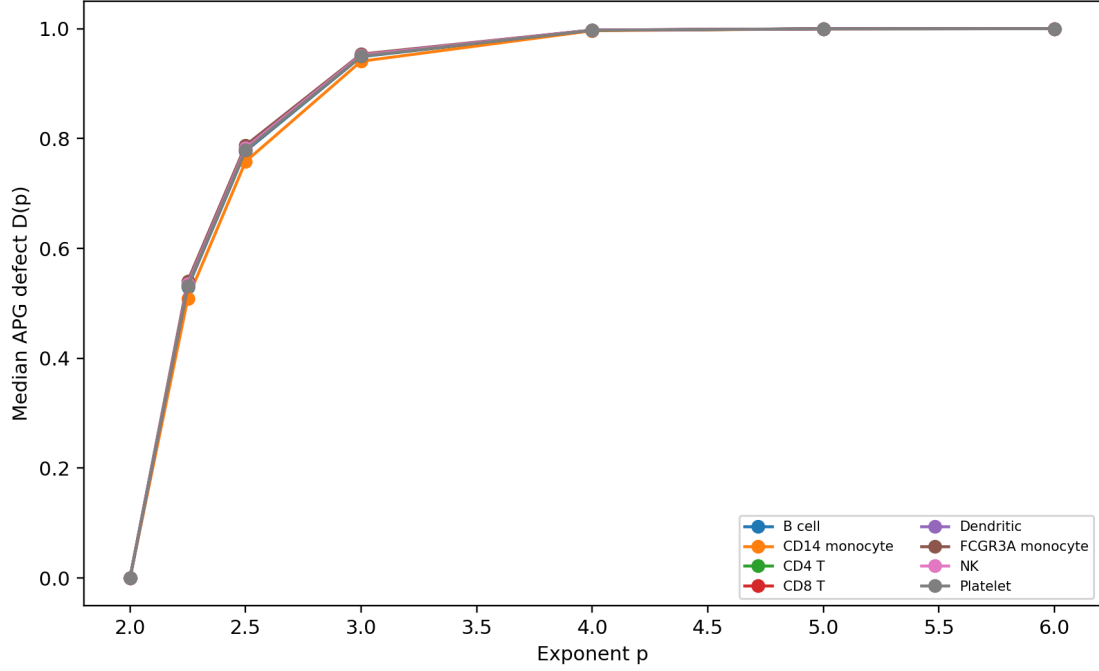


Figure 3: Median gene-level APG deformation profiles by provisional class. Curves rise rapidly between $p = 2$ and $p = 3$ and then approach saturation.

Table 6: Three-fold stratified internal classification results on a balanced 6,400-cell subset. Values are mean $\pm$ standard deviation across identical folds.

Representation	Features	Balanced accuracy	Macro-F1
M1: Shannon only	1	0.2189 $\pm$ 0.0040	0.1710 $\pm$ 0.0057
M2: Classical diversity	7	0.3625 $\pm$ 0.0059	0.3027 $\pm$ 0.0058
M3: Classical + gene APG	18	0.4033 $\pm$ 0.0086	0.3792 $\pm$ 0.0104
M4: Programme APG	11	0.4183 $\pm$ 0.0053	0.3887 $\pm$ 0.0034
M5: Classical + gene and programme APG	27	0.5400 $\pm$ 0.0020	0.5299 $\pm$ 0.0037

Shannon entropy was the most stable tested descriptor at every depth, retaining $\rho = 0.951$ at 20%. High-exponent $D(6)$ and maximum concentration were more sensitive, with $\rho = 0.770$ and 0.803 at 20%. Thus, APG deformation was robust to substantial thinning, but it was not uniformly more robust than classical entropy.

5 Discussion

This study establishes a complete computational benchmark for applying APG to sparse single-cell transcriptomes. The principal positive result is not that APG replaces existing single-cell methods. Rather, APG supplies a controlled family of bounded deformation measurements that can be evaluated alongside entropy, concentration, and effective dimension. At both gene and programme levels, the APG descriptors added internal discriminatory information beyond the classical compact summaries tested here.

The distinction between gene-level and programme-level results is informative. Gene-level expression spans more than thirty thousand coordinates and the higher-exponent defects saturate rapidly. Programme aggregation reduced the representation to nine interpretable axes and

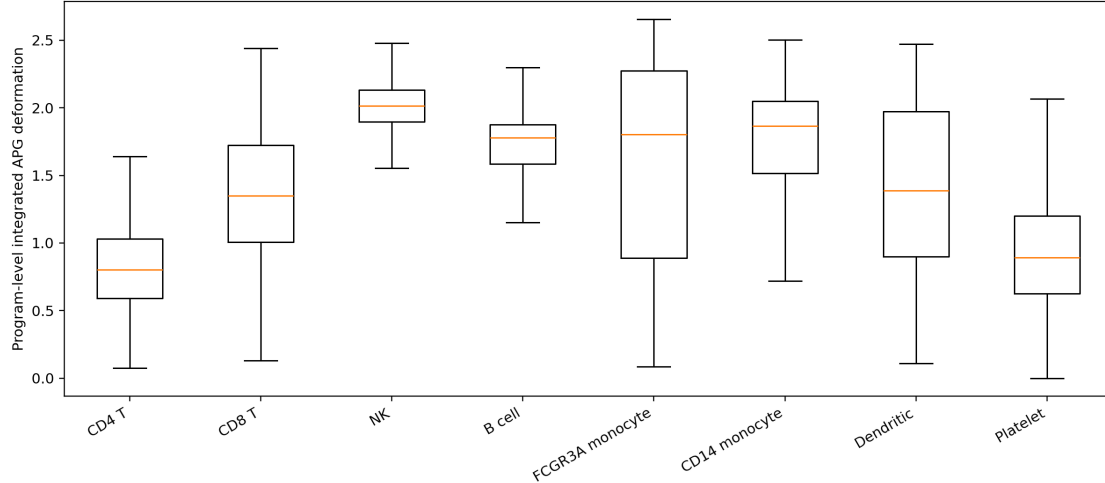


Figure 4: Programme-level integrated APG deformation from $p = 2$ to $p = 6$ across provisional PBMC classes.

Table 7: Euclidean silhouette coefficients in standardized descriptor spaces.

Representation	Cells	Silhouette
M5: Classical + gene and programme APG	1,500	-0.096
M3: Classical + gene APG	1,500	-0.143
M1: Shannon only	1,500	-0.152
M2: Classical diversity	1,500	-0.158
M4: Programme APG	1,500	-0.214

produced substantially larger between-class effect sizes. This supports the use of APG after biologically meaningful aggregation, where concentration and deformation can be associated with named cellular programmes. At the same time, current single-cell pathway-scoring benchmarks show that programme scores are sensitive to gene-set design and sparsity (Wang et al., 2024). The present mean-expression modules therefore provide a transparent starting point, not a definitive pathway-scoring standard.

The classification comparison offers evidence of incremental information, but its scope must be read carefully. The targets were generated by marker modules, and several programme modules contained genes related to the same immune identities. Consequently, the M5 result measures consistency within a marker-informed representation and may include circularity. A stronger future design must obtain reference annotations independently, freeze all marker and programme definitions before evaluation, and test the fitted representation on donors and laboratories not used in construction. Modern single-cell integration and probabilistic models are explicitly designed for these multi-sample settings (Stuart et al., 2019; Hao et al., 2021; Lopez et al., 2018); APG should be evaluated as a compact interpretable layer attached to such workflows rather than as a substitute for them.

The negative silhouettes reveal a second limitation. Although supervised models extracted useful decision boundaries, none of the compact spaces formed well-separated Euclidean clusters. This argues against presenting the current descriptor set as a universal cell-state embedding. Its more defensible role is a structural summary: a small vector of quantities that can be tracked, compared, and interpreted alongside conventional embeddings.

The downsampling study showed that APG entropy and integrated deformation were stable

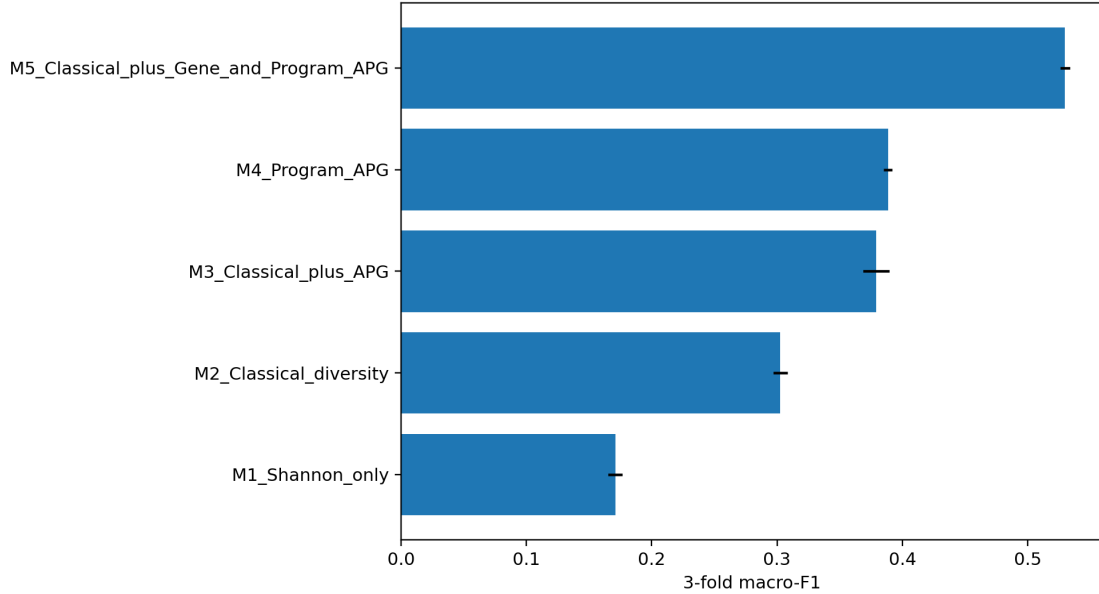


Figure 5: Macro-F1 comparison for the five compact representations under identical three-fold stratified splits. Error bars show the standard deviation across folds.

Table 8: Spearman rank correlation between full-depth and thinned-count descriptors on 3,000 cells.

Retained	Shannon	APG entropy	$C_{\max}$	$D(3)$	$D(6)$	$A(2, 6)$
80%	0.996	0.994	0.978	0.992	0.987	0.993
60%	0.992	0.986	0.943	0.978	0.958	0.981
40%	0.981	0.966	0.890	0.946	0.889	0.954
20%	0.951	0.917	0.803	0.878	0.770	0.893

even after aggressive count reduction. However, Shannon entropy was more stable than all APG quantities tested, and high-exponent descriptors were particularly sensitive. This pattern is mathematically plausible because squaring and high powers emphasize dominant coordinates and therefore amplify count perturbations. Low to moderate exponents should be preferred when sequencing depth is limited, while $D(6)$ may be useful mainly as a sensitive concentration descriptor in sufficiently deep data.

Several limitations remain. First, the dataset contains one donor, preventing donor-level inference or clinical generalization. Second, the filtered matrix was accepted as supplied; raw-droplet cell calling, ambient RNA correction, doublet detection, and donor-specific quality-control sensitivity were not repeated. Third, the labels are provisional and not expert-validated. Fourth, the programme collection is small and manually curated. Fifth, the squared APG mapping increases the influence of highly expressed genes and requires comparison with linear-weight alternatives. Sixth, all inferential p values are inflated by the large number of cells and should not be interpreted as independent biological replication. Seventh, the internal folds divide cells from the same donor, so their variance reflects cell-level resampling rather than cohort generalization.

A multi-donor extension should use independent reference labels, donor-held-out validation, curated pathway databases, gene-set scoring methods such as UCell or AUCCell, explicit comparison with PCA and latent models, and mixed-effects or pseudobulk inference where condition-level claims are made. The present package provides the mathematical definitions, executable code, complete cell-level descriptors, figures, and tables required for that next stage.

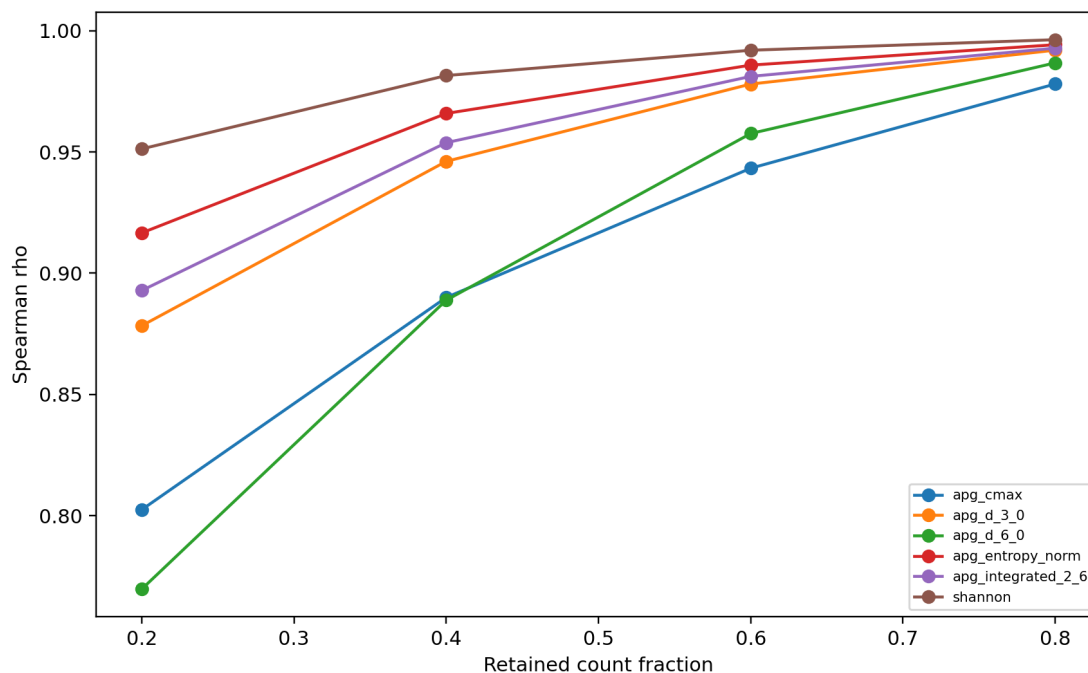


Figure 6: Descriptor rank stability under binomial thinning of raw UMI counts. Curves show Spearman correlation with full-depth values.

6 Conclusion

APG can be computed at the scale of 68,579 single-cell transcriptomes and compared transparently with established diversity and concentration measures. Gene-level APG added internal discrimination beyond classical compact descriptors, while programme-level APG produced the strongest between-class effects and improved the combined classification representation. The results also identify clear boundaries: the descriptor spaces were not naturally cluster-separating, high-exponent quantities were less robust under severe thinning, the labels were marker-derived, and the dataset contained only one donor. APG is therefore supported as an interpretable complementary structural representation for exploratory single-cell analysis, not as a validated universal cell-state space or clinical classifier.

Data availability

The analyses were performed using the publicly available *Fresh 68k PBMCs (Donor A)* filtered gene-cell matrix from the official 10x Genomics dataset repository (<https://www.10xgenomics.com/datasets/fresh-68-k-pbm-cs-donor-a-1-standard-1-1-0>). The original source matrix is not redistributed. The reproducibility package includes the Cell Ranger summary report, derived APG descriptors, figures, tables, and supplementary materials.

Code availability

The complete analysis pipeline, source code, figures, tables, and reproducibility package are available at: <https://github.com/Arithmetic-Power-Geometry/APG-PBMC-68K-Analysis>

An archived, citable version is available at: <https://doi.org/10.5281/zenodo.21306239>

Ethics statement

The study is a secondary computational analysis of a publicly released, de-identified benchmark dataset and involved no new recruitment, intervention, or collection of identifiable human data.

Competing interests

The author declares no competing interests.

Author contributions

M.A.K.A. conceived the APG single-cell framework, designed the study, implemented the analysis, interpreted the results, and wrote the manuscript.

Funding

No external funding was received for this study.

References

- Abdelaal, T., Michielsen, L., Cats, D., Hoogduin, D., Mei, H., Reinders, M. J. T., and Mahfouz, A. (2019). A comparison of automatic cell identification methods for single-cell rna sequencing data. *Genome Biology*, 20:194.
- Aibar, S., González-Blas, C. B., Moerman, T., Huynh-Thu, V. A., Imrichova, H., Hulselmans, G., Rambow, F., Marine, J.-C., Geurts, P., Aerts, J., van den Oord, J., Atak, Z. K., Wouters, J., and Aerts, S. (2017). Scenic: Single-cell regulatory network inference and clustering. *Nature Methods*, 14:1083–1086.
- Akhtar, M. A. K. (2026a). Foundations of arithmetic power geometry: Local parameter deformations and entropy at the euclidean target. Zenodo. <https://doi.org/10.5281/zenodo.20747768>.
- Akhtar, M. A. K. (2026b). Shannon–rényi control of the arithmetic power geometry closure defect. Zenodo. <https://doi.org/10.5281/zenodo.20749524>.
- Andreatta, M. and Carmona, S. J. (2021). Ucell: Robust and scalable single-cell gene signature scoring. *Computational and Structural Biotechnology Journal*, 19:3796–3798.
- Aran, D., Looney, A. P., Liu, L., Wu, E., Fong, V., Hsu, A., Chak, S., Naikawadi, R. P., Wolters, P. J., Abate, A. R., Butte, A. J., and Bhattacharya, M. (2019). Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nature Immunology*, 20:163–172.
- Gini, C. (1912). Variabilità e mutabilità. *Studi Economico-Giuridici della R. Università di Cagliari*, 3:1–158.
- Hao, Y., Hao, S., Andersen-Nissen, E., Mauck, W. M., Zheng, S., Butler, A., Lee, M. J., Wilk, A. J., Darby, C., Zager, M., et al. (2021). Integrated analysis of multimodal single-cell data. *Cell*, 184(13):3573–3587.e29.

- Heumos, L., Schaar, A. C., Lance, C., Litinetskaya, A., Drost, F., Zappia, L., Lücken, M. D., Strobl, D. C., Henao, J., Curion, F., et al. (2023). Best practices for single-cell analysis across modalities. *Nature Reviews Genetics*, 24:550–572.
- Kruskal, W. H. and Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association*, 47(260):583–621.
- Lloyd, M. and Ghelardi, R. J. (1964). A table for calculating the “equitability” component of species diversity. *Journal of Animal Ecology*, 33(2):217–225.
- Lopez, R., Regier, J., Cole, M. B., Jordan, M. I., and Yosef, N. (2018). Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15:1053–1058.
- Luecken, M. D. and Theis, F. J. (2019). Current best practices in single-cell rna-seq analysis: A tutorial. *Molecular Systems Biology*, 15(6):e8746.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, 12:2825–2830.
- Pullin, J. M. et al. (2024). A comparison of marker gene selection methods for single-cell rna sequencing data. *Genome Biology*, 25:48.
- Rényi, A. (1961). On measures of entropy. In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 547–561. University of California Press.
- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65.
- Shannon, C. E. (1948). A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656.
- Simpson, E. H. (1949). Measurement of diversity. *Nature*, 163:688.
- Stuart, T., Butler, A., Hoffman, P., Hafemeister, C., Papalexi, E., Mauck, W. M., Hao, Y., Stoeckius, M., Smibert, P., and Satija, R. (2019). Comprehensive integration of single-cell data. *Cell*, 177(7):1888–1902.e21.
- Wang, R. H. et al. (2024). Comparative analysis of single-cell pathway scoring methods. *NAR Genomics and Bioinformatics*, 6(3):lqae124.
- Wolf, F. A., Angerer, P., and Theis, F. J. (2018). Scanpy: Large-scale single-cell gene expression data analysis. *Genome Biology*, 19:15.
- Zheng, G. X. Y., Terry, J. M., Belgrader, P., Ryvkin, P., Bent, Z. W., Wilson, R., Ziraldo, S. B., Wheeler, T. D., McDermott, G. P., Zhu, J., Gregory, M. T., Shuga, J., Montesclaros, L., Underwood, J. G., Masquelier, D. A., Nishimura, S. Y., Schnall-Levin, M., Wyatt, P. W., Hindson, C. M., Bharadwaj, R., Wong, A., Ness, K. D., Beppu, L. W., Deeg, H. J., McFarland, C., Loeb, K. R., Valente, W. J., Ericson, N. G., Stevens, E. A., Radich, J. P., Mikkelsen, T. S., Hindson, B. J., and Bielas, J. H. (2017). Massively parallel digital transcriptional profiling of single cells. *Nature Communications*, 8:14049.

A Supplementary programme definitions

Table 9: Curated programmes and genes used for programme-level APG.

Programme	Genes
T-cell receptor signalling	CD3D, CD3E, CD3G, TRAC, LCK, LAT, ZAP70, CD247
Cytotoxicity	NKG7, GNLY, PRF1, GZMB, GZMH, CTSW, CCL5
B-cell receptor activity	MS4A1, CD79A, CD79B, CD37, CD74, HLA-DRA
Interferon response	ISG15, IFIT1, IFIT2, IFIT3, IFI6, MX1, OAS1, STAT1
Inflammatory myeloid	LYZ, S100A8, S100A9, FCN1, CTSS, TYROBP, LST1, LGALS3
Oxidative phosphorylation	NDUFA1, NDUFB1, NDUFS5, COX5A, COX6A1, ATP5F1, ATP5G1, UQCRCB
Ribosomal activity	RPL3, RPL13, RPL32, RPS3A, RPS6, RPS12, RPS18, RPLP0
Cell cycle	MKI67, PCNA, STMN1, TUBA1B, TYMS, MCM2, MCM5, TOP2A
Apoptosis/stress	BAX, BCL2, FOS, JUN, DUSP1, HSPA1A, HSP90AA1, IER2

B Supplementary marker definitions

Table 10: Marker modules used for provisional cell annotation.

Class	Marker genes
CD4 T	IL7R, LTB, MAL, CCR7, LDHB
CD8 T	CD8A, CD8B, CCL5, GZMK
NK	NKG7, GNLY, PRF1, GZMB, KLRD1, FGFBP2
B cell	MS4A1, CD79A, CD79B, CD74, CD37, HLA-DRA
CD14 monocyte	LYZ, S100A8, S100A9, CTSS, FCN1, CD14, TYROBP
FCGR3A monocyte	FCGR3A, MS4A7, LST1, IFITM3, SERPINA1, LGALS3
Dendritic	FCER1A, CST3, CLEC10A, CD1C, HLA-DPA1, CLEC4C
Platelet	PPBP, PF4, NCOA4, GNG11, SDPR, ITGA2B